

Your grade: 100%

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Next item →

1. What are possible reasons for a large entry in the Kalman gain matrix L_k ? Select all that apply.

1 / 1 point

☒ The corresponding state is highly coupled to the corresponding output, so the measurement contains a lot of information that could be helpful in updating the state.

✔ **Correct**
Yes. This is one possible reason.

☒ We are very confident in the predicted value of the state $\hat{z}_k$.

✔ **Correct**
Yes. This is one possible reason (meaning that the innovation has quality information for updating this state value).

☒ The corresponding state has a high uncertainty and could benefit from a large adjustment.

✔ **Correct**
Yes. This is one possible reason.

☐ The corresponding measurement has a lot of sensor noise, which will be suppressed by this large entry in L_k .

2. To be able to compute $\hat{x}_k^+$ and $\Sigma_{\hat{x},k}^+$, what do we need to know? Select all that apply.

1 / 1 point

☒ The state prediction error covariance $\Sigma_{\hat{x},k}^-$.

✔ **Correct**
Yes. This is needed by steps 2a and 2c.

☐ The process-noise covariance matrix $\Sigma_{\bar{w}}$.

☒ The sensor-noise covariance matrix $\Sigma_{\bar{v}}$.

✔ **Correct**
Yes. This is needed by step 2a.

☒ The predicted value of the state at the present time, $\hat{x}_k^-$.

✔ **Correct**
yes. This is needed by step 2b.

☒ The model matrices C_k and D_k .

✔ **Correct**
Yes. These are needed by steps 2a and 2c.

3. What is the significance of the state estimation error covariance matrix $\Sigma_{\hat{x},k}^+$?

1 / 1 point

☐ It indicates the uncertainty of the predicted state.

☒ It allows computing error bounds on the state estimate produced by the Kalman filter.

☐ It is needed to compute the output prediction error $\tilde{z}_k = z_k - \hat{z}_k$.

☐ It is needed to be able to compute the Kalman gain matrix L_k .

✔ **Correct**
Yes. This is the primary significance and use of this matrix.